



RKDF UNIVERSITY, BHOPAL

B.Sc.(Zoology)

SCHEME

Semester – I to VI

No	Subject Code	Subject Title	Marks Allotted						
			Assignment Marks		Theory Marks		Practical Marks		Total Marks
			Max	Min	Max	Min	Max	Min	
1	BSZ 111	INVERTEBBRATA & CHORDATA	20	8	80	27	50	17	150
2	BSZ 121	CELL BIOLOGY & GENETICS	20	8	80	27	50	17	150
3	BSZ 131	ANIMAL PHYSIOLOGY & DEVELOPMENTAL BIOLOGY AND IMMUNOLOGY	20	8	80	27	50	17	150
4	BSZ 141	ECOLOGY & EVOLUTIONARY BIOLOGY	20	8	80	27	50	17	150
5	BSZ 151	MICROBIOLOGY AND BIOCHEMISTRY	20	8	80	27	50	17	150
6	BSZ 161	BIostatISTICS AND COMPUTER APPLICATIONS & BASIC ANIMAL BIOTECHNOLOGY	20	8	80	27	50	17	150



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Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Zoology)	INVERTEBRATA & CHORDATA	BSZ 111

Unit-1

A brief introduction and nomenclature - Level of Organization. Phylum Protozoa: General characters- Classifications. Type study-Paramecium-Structure and Reproduction. General Topic: Protozoan Diseases.

Unit -2

Phylum Porifera: General characters- Type study- Ascon- Cellular structure. Phylum Coelenterata: Classification- Type study- Aurelia- Structure and life history. General Topic: Canal system in Sponges, Polymorphism in Coelenterates. Phylum Mollusca: General characters- Type study – Unio (Lamellidens) - External morphology and digestive system. General Topic: Mouth parts of Insects- Economic Importance of Mollusca.

Unit -3

Phylum Platyhelminthes: General Characters- Classification- Type study- Liver fluke- Structure and Reproduction. Phylum Annelida: General characters- Type study- Nereis- External morphology and reproduction. General Topic: Helminth Parasites in Man.

Unit-4

Introduction- Type Study: Amphioxus- external characters, digestive, excretory, respiratory, and circulatory systems.

Class: Pisces, General characters – Type Study: Scoliodon-External characters, Digestive, Excretory, Respiratory and Circulatory Systems- Structure of Brain-Sense organs and Reproductive system. General Topic: Accessory respiratory organs in fishes.

Unit-5

Class : Amphibia :General characters and classification -Type Study : Frog –External characters, Digestive ,Respiratory, Circulatory and Reproductive systems - Structure of Brain.

Class: Reptelia: General characters- Type study –Calotes- External characters-Digestive, Respiratory, Circulatory and Reproductive systems- Structure of Brain. General Topic: Identification of Poisonous and Non- Poisonous snakes. Golden age of Reptiles.

References

1. Agarwal V.K (2000) Invertebrate Zoology- S.Chand Company.
2. Barnes R.D (1987) Invertebrate Zoology- Saunders College Publications.
3. Barrington E.J (1981) Invertebrate Structure and Function. ELBS Editions.
4. Ekambaranatha Iyer (1993) Manual of Zoology Volume I Invertebrata.
5. Kotpal R.L (2003) Modern text book of Zoology- Rostogi Publications, Meerut.
6. Ekambaranatha Iyer (1993) Manual of Zoology Vol.II, Viswanathan (printers& publishers) Chennai.
7. Jordon, E.L & Verma, P.S. (2000) Chordate Zoology, S.Chand & Co, New Delhi.
8. Newman H.H., Chordata, McMillan publishers.



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SEMESTER-I

PRACTICE-I

INVERTEBRATA AND CHORDATA

I. Major Practicals:

Cockroach-Nervous, digestive, Reproductive system

Prawn-Nervous system

II. Minor Practicals:

Prawn –Appendages

Mouth parts –Honey Bee, Mosquito, and Cockroach.

III. Spotters:

a) Classify giving reactions:

Entamoeba, Paramecium, Leucosolenia, Hyalonema, Aurelia, Obelia, Taenia, Ascaris, Earthworm, Nereis, Penaeus, Freshwater mussel, Starfish, Cockroach, Amphioxus, Salpa, Frog, Cobra, Pigeon, Rabbit.

b) Draw Labelled Sketch:

T.S. of Taenia, T.S. of Fasciola, Ephyra larva, Nauplius larva, Zoea larva, Quill feather, Frog – Pectoral girdle, Pigeon –Pelvic girdle.

c) Biological Significance:

Sponge –Gemmule, Physalia, Leech, Limulus, Bipinnaria, Ascidian tadpole larva, Ichthyophis, Peripatus.

d) Relate structure and function

Taenia –Scolex, Nereis – Parapodium, Penaeus –Pereopods, Star fish –Tube feet (ventral view), Echinoderms, Draco, Bat.

e) Comment on Respiratory /skeletal/ dentition of the following

Star fish, Synsacrum, Dentition of Rabbit and Dog.



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Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Zoology)	CELL BIOLOGY & GENETICS	BSZ 121

UNIT 1

Prokaryotic and eukaryotic cells –Ultrastructure and Organization. Plasma membrane–Ultra structure–Chemical composition and functions of modifications of plasma membrane. Endoplasmic reticulum: Morphology, Ultra structure, chemical composition and functions. Golgi complex: Ultra structure, chemical composition and functions.

UNIT 2

Lysosomes: Ultra structure and polymorphism- chemical composition and functions: Peroxisomes and glyoxysomes. Mitochondria: Ultra structure- chemical composition-enzyme systems-functions-Oxidation- Respiratory chain (ETP)- Kreb's cycle, ATP Production and Biogenesis.

UNIT 3

Ribosomes: Ultra structure-types- chemical composition - functions. Nucleus and Neucleolus: Ultra structure of Nucleus and Nucleolus. Nucleic Acids: DNA –Ultra structure-replication-transcription, Chromosomes: Ultra structure of Chromosomes and Giant Chromosomes, Cell division- mitosis and meiosis. Cell cycle,

UNIT 4

Introduction –Laws of Mendel –Interaction of genes (Epistatic gene ,Complementary genes and Lethal genes . Inheritance of Blood group in man and Coat colour in Rabbit . Mechanism of linkage and crossing over –Types and theories - Chromosomal Mapping –Sex linked inheritance (haemophilia, and colour blindness). Sex limited inheritance and sex influenced inheritance .

UNIT 5

Sex determination in man, Drosophila and Bonellia. Mutations – Point mutation and Chromosomal aberrations and mutagens . Inbreeding and out breeding, heterosis –Genetic applications in animals. DNA as genetic material –Experiments . Syndromes (Down syndrome and Turners syndrome in man) .

TEXT BOOKS:

Cell biology. Veer Bala Rastogi, Rastogi Publications.

Cell Biology, Power.Verma P.S. and Agarwal V. K . –Concepts of Genetics .

Rastogi V.B. A text book of Genetics, K.Ramnath, Meerut.



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YEAR-I, SEMESTER-II PRACTICAL CELL BIOLOGY AND GENETICS

A. Major Practicals

Use Microscopes, Camera Lucida, Stage and Ocular micrometers.

Total Counting of RBC / WBC Using haemocytometer.

Blood Smear Preparation, Differential count of WBC.

Mounting Buccal Epithelium and observing living cells using vital staining.

Study of mitotic division using onion root tips.

Study of prepared slides of different tissues.

Submission of practical record.

B.GENETICS PRACTICALS

Observation of common mutants of drosophila

Preparation of mounting of the salivary gland in chironomous larva

Submission of practical record.



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Syllabus Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Zoology)	ANIMAL PHYSIOLOGY & DEVELOPMENTAL BIOLOGY AND IMMUNOLOGY	BSZ 131

Unit-1

Nutrition –types, Enzymes – Enzyme action, Coenzymes, Digestion in man. Respiration – Respiratory pigments, role in transport of O₂ and CO₂ in man. Circulation - blood composition, origin and conduction of heart beat in an – blood pressure, Heart diseases– heart attack.

Unit-2

Excretion – types of nitrogenous wastes – structure of the mammalian kidney and urine formation – renal failure – kidney stone – kidney transplantation. Osmotic – ionic regulation in freshwater, marine, estuarine and terrestrial organisms

Unit-3

Amoeboid, ciliary and flagellar movements. Types of muscles – ultra structure of skeletal muscle – Muscle contraction and theories. Neuron, - Types- Impulse transmission, -synaptic transmission - reflex action. -Endocrine glands in man, secretions and disorders.

Unit-4

Spermatogenesis – definition – process and significance, structure of mammalian sperm. Oogenesis – definition – process and significance – Types of eggs and egg membranes. Fertilization – definition – process and significance. Parthenogenesis – definition and significance – types of parthenogenesis.

Unit-5

Cleavage patterns (types) – Cleavage in Frog, Chick and Mammals. Morula and Blastulation. Introduction – cells and organs involved in immune response. Types of immunity – Innate & adaptive immunity of acquired immunity, humoral and cell mediated immunity, active and passive immunity.

TEXT BOOKS:

Verma P.S. & Tyagi B.S. Animal Physiology, 6th edition. S.Chand & Co.
Agarwal, V.K. Agarwal, R.A.Srivastava A.K. & Kausha Kumar, Animal physiology & Biochemistry, S. Chand & Co.,
De Beer, G.R. Embryos and Ancestors. Clarendon Press, Oxford.
Verma. P.S and Agarwal, V.K. Chordate Embryology, S.Chand and Co. Ltd., New Delhi (1998).
Bodmer, Modern Embryology, Saunders International student edition, Philadelphia.3 rd Edition 1981.
Eli Benjamini et al.,(1991) Immunology – A short course – Wiley Publishers, NY.

REFERENCES:

Hoar, W.S (1987) General and Comparative physiology, prentice – Hall.
M.K.Chandrashekar – Circadian Rhythms – Madras science foundation, Chennai.



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YEAR-II SEMESTER-III

PRACTICAL

(ANIMAL PHYSIOLOGY, AND DEVELOPMENTAL BIOLOGY)

I Major Practicals:

- 1) Qualitative analysis of digestive enzymes in cockroach.
- 2) Estimation of the rate of O_2 consumption in fish/crab with reference to body weight.
- 3) Detection nitrogenous waste products in fish tank water, bird excreta & mammalian urine.
- 4) Study of human salivary activity in relation to temperature.
- 5) Qualitative analysis of carbohydrates, proteins, and amino acids.

II Minor Practicals:

Kymograph –simple twitch, Trappe, Fatigue, Tetanus, Spigmanometer, pH meter, Colorimeter, Haemometer, Enzyme action – graphs (temperature, concentration of substrate and enzyme.)



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Syllabus

Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Zoology)	ECOLOGY & EVOLUTIONARY BIOLOGY	BSZ 141

UNIT I

Abiotic factors of the Environment : Temperature, Light and .Oxygen. Bio geo chemical cycles with special reference to Nitrogen Phosphorous and Carbon. Biotic factors of the environment - Animal relationship.

UNIT II

Population : characteristics –Natality, Mortality, Density, and age distribution, population control, life-tables, Food chains, Food webs and Ecological pyramids. Air pollution, Water pollution and Oil pollution. Noise pollution and Thermal pollution.

UNIT III

Pond as a Ecosystem, energy flow and ecological succession. Habitats – Terrestrial – Aquatic – Marine, Fresh water and estuary. Environmental resources- renewable and non renewable resources. Forest resources- Protection – Chipko movement- A forestation. Wild life management- Wild life sanctuaries and National Parks.

Unit-IV

History of Evolutionary thought - Origin of life –Chemical evolution. Evolution of self replicating systems –DNA world and RNA world.

Evidences from Paleontology -Comparative anatomy, Embryology, Physiology and Bio chemistry. Bio geography –Distribution in continents ,Continuous and discontinuous distribution - Endemism .

Unit-V

Natural selection- Species and Speciation- Sympatric and allopatric speciation. Isolating mechanism- mutation and genetic drift.

Adaptation and adaptive radiation, Colouration-mimicry-Darwins finches. Polymorphism-types and significance. Convergent-Divergent-parallel and co-evolution of Man and cultural evolution.

TEXT BOOKS

1. H.D.Kumar, Modern concepts of Ecology. Vikas Publishing house.
2. E.P. Odum, Fundamentals of Ecology.
3. G.C. Clarke, Elements of Ecology, John Wiley sons, New York
4. Rostogi, V.B. Organic Evolution, Kedernath, Ramnath publishers, Meerut.
- 5, Verma P.S. & Agarval, V.L. concepts of evolution S.Chand& Company.

REFERENCES:

Introduction to evolution-Dodson-Evolution: process and product.



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PRACTICAL

Spotters:

Description and uses of autoclave, Hot air oven, Incubator, Water bath, Centrifuge, Refrigerator, pH meter, Colorimeter, Microtome, Rain gauge, Anemometer, Maximum minimum thermometer, Hygrometer, and Barometer.

Computer applications - Hardware of computer, storage device, mouse.

Submission of field Report.

Submission of Practical Record.



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Syllabus Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Zoology)	MICROBIOLOGY AND BIOCHEMISTRY	BSZ 151

Unit-1

Classification of microorganisms- General characteristics of Bacteria, Virus, Yeast. Bacteria- Morphology, Bacterial cell structure, Motility, Nutrition and Reproduction. Virus-discovery- Morphology, Classification, phages and life cycle. Yeast-Morphology, cell structure, Multiplication, phages and cycle.

Unit-2

Morphology of water, air soil and sewage. Water-Microorganisms of water, total bacterial count. Air-Microorganisms in soil, nitrogen cycle. Sewage-Composition of sewage, treatment of sewage by microorganisms.

Unit-3

Food borne diseases- Microbial food poisoning by Salmonella and Clostridium botulinum (Botulism). Measures to prevent microbial food poisoning. Food infection-Food borne diseases- Diarrhea, Dysentery, Typhoid and Cholera. Water borne diseases-Hepatitis, Gastro enteritis, Campylobacter-diarrhea, Giardia lamblia, Cryptosporidiosis cholera. Air borne diseases-Common cold, Tuberculosis, Pneumonia, Diphtheria.

Unit-4

Biochemistry- Definition and its importance, Physio- chemical forces acting on the living body –a) Definition of pH and its determination, Maintenance of pH of blood.
b) (Definition of osmosis, abnormality in edema and dehydration.
Nucleic acids, structure and classification.

Unit-5

Carbohydrates, lipids, Amino acids & proteins-Classification, structure and their function. Metabolism- Glycolysis-TCA cycle - Electron Transport chain, Urea cycle Deamination, Oxidation of fatty acids.

REFERENCES:

Microbiology- Pelzer.
Biology of Microorganism- Madigan-Brock
Microbiology Lab manual – Capachin.
Microbiology fundamentals and application- Atlas.R.M.
Principles of Biochemistry A.L. Lehninger, D.L. Nelson & M.M. Cox (1993) Worth publishes New York.
Biochemistry by L.Stryer (1994) freeman & co., Newyork.
Biochemistry by Zubay (1998) Macmillan publishers & co., New York.



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MAJOR PRACTICALS:

1. Estimation of dissolved oxygen content in the given water sample (Winklers method).
2. Estimation of salinity and pH in given water sample.
3. Plankton study –Identification and description of any five marine planktons.

MINOR PRACTICALS:

4. Examination of yeast, mould, protozoa and pathogenic bacteria.
5. Estimation of urine sugar.
6. Blood grouping.
7. Problems on calculation of Mean, median, mode.

Spotters

Developmental Biology-Slides

Slides of mammalian sperm and Ovum

Slides of different developmental stages of chick embryos (24, 48, 72, 96 hrs)

Slides of blastula and gastrula of frog (morula, early gastrula, yolk plug stage, late gastrula)

Placenta of Sheep / Pig/ Rat.



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Syllabus Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc(Zoology)	BIostatISTICS AND COMPUTER APPLICATIONS & BASIC ANIMAL BIOTECHNOLOGY	BSZ 161

Unit-1

Introduction-definition, data types – primary and secondary – Classification of data, Collection of data – tabular and graphical representation – Bar diagram, Pi diagram, Column graph, Histogram, Ogive curves.

Unit-2

Measures of central tendency – Mean, Mode and Median, Variance, Standard deviation, Standard error and Coefficient of variance. Simple Correlation, Simple Regression, Chi square test, student's – t- test.

Unit-3

Fundamentals of Computer: Classification, Computer organization, Input devices, processing unit, output devices, external storage devices, software, WWW, CONCEPT OF E-Mail.

Computer and its application to biology-Definition and scope of Bioinformatics - application and introduction to Biological data.

Unit-4

History of animal cell culture, Laboratory requirements for animal cell culture, Sterilization techniques.

Media used for animal cell culture, Types of cell culture (Primary and Secondary), Introduction to established cell lines, Stem cells.

Unit-5

Cryopreservation of cell cultures, application of animal cell cultures in production of therapeutic proteins. Hybridoma technology.

Gene transfer methods in Animals – Microinjection, Embryonic Stem cell gene transfer, Retrovirus & Gene transfer. Transgenic Animals, Animal propagation – Artificial insemination,

REFERENCES:

Introduction of Biostatistics and Computer Science – Y.I Parkar & M.G Dhanyagude NiraliPrakashan publishers, Pune.

Biostatistics by K.S. Negi ATIBS publications & distributors, New Delhi.

Bishop O.N. Statistics for Biology. Boston, Hollghtan, Mifflin.

Introduction to Biostatistics by Pranab kumar, S.Chand company Ltd. New Delhi.